DESCENT OF M-PSEUDO-COHERENCE ALONG FPQC COVERINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Pullback preserves m-pseudo-coherence, by the cohomology pseudo-coherent pullback lemma. invoked at proof lines 3053-3054. Gives the forward implication.
- An fpqc covering of an affine open can be refined by finitely many affine opens of the members mapping into it, so the converse reduces to a single surjective flat morphism of affine schemes. invoked at proof lines 3056-3063. Sets up the affine descent situation.
- Over a ring, m-pseudo-coherence descends along faithfully flat ring maps, by the More on Algebra flat descent lemma, applied through the affine comparison and the pseudo-coherent affine lemmas. invoked at proof lines 3064-3067. Concludes the converse.

Source reference. Original lemma 09U7: sources/stacks-chow/tags.txt, source lines 12415–12415. Cited at proof lines 3053-3054 for the forward implication; the descent argument is inside the counted span.

Source reference. Original lemma 068R: sources/stacks-chow/tags.txt, source lines 8015–8015. Cited at proof lines 3065-3066 to conclude in the affine case; the reduction to it is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 3065-3066 to translate the statement into modules; the descent conclusion is inside the counted span.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Cited at proof lines 3066-3067 together with the comparison lemma; the descent conclusion is inside the counted span.

Lemma 0.2. *Let $\{f_i : X_i \to X\}$ be an fpqc covering of schemes. Let $E \in D_{QCoh}(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$. Then E is m-pseudo-coherent if and only if each Lf_i^*E is m-pseudo-coherent.*

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Proof. Pullback always preserves m-pseudo-coherence, see Cohomology, Lemma 09U7. Conversely, assume that Lf_i^*E is m-pseudo-coherent for all i. Let $U \subset X$ be an affine open. It suffices to prove that $E|_U$ is m-pseudo-coherent. Since $\{f_i : X_i \to X\}$ is an fpqc covering, we can find finitely many affine open $V_j \subset X_{a(j)}$ such that $f_{a(j)}(V_j) \subset U$ and $U = \bigcup f_{a(j)}(V_j)$. Set $V = \coprod V_i$. Thus we may replace X by U and $\{f_i : X_i \to X\}$ by $\{V \to U\}$ and assume that X is affine and our covering is given by a single surjective flat morphism $\{f : Y \to X\}$ of affine schemes. In this case the result follows from More on Algebra, Lemma 068R via Lemmas 06Z0 and 08E7. $\square$